DATA STRUCTURE

LAB PROGRAMS

8. a) Write a C program for deletion of a newnode at the beginning of a single linked list .

```
#include <stdio.h>

#include <stdlib.h>

struct Node

{

   int data;

   struct Node *next;

};

void display(struct Node *head)

{

   struct Node *temp = head;

   while(temp != NULL)

   {

      printf("%d -> ", temp->data);

      temp = temp->next;

   }

   printf("NULL\n");

}

int main()

{

   struct Node *head, *temp;

   head = (struct Node *)malloc(sizeof(struct Node));

   head->data = 10;
```

```
    head->next = (struct Node *)malloc(sizeof(struct Node));
    head->next->data = 20;
    head->next->next = (struct Node *)malloc(sizeof(struct Node));
    head->next->next->data = 30;
    head->next->next->next = NULL;
    printf("Before Deletion:\n");
    display(head);
    head = head->next;
    free(temp);
    printf("\nAfter Deletion at Beginning:\n");
    display(head);
    return 0;
}
```

Output

```
Before Deletion:
10 -> 20 -> 30 -> NULL

After Deletion at Beginning:
20 -> 30 -> NULL


=== Code Execution Successful ===
```

b) Write a C program for deletion of a newnode at the end of a single linked list .

```
#include <stdio.h>
#include <stdlib.h>
struct Node
{
    int data;
    struct Node *next;
};
void display(struct Node *head)
{
    struct Node *temp = head;
    while(temp != NULL)
    {
        printf("%d -> ", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
}
int main()
{
    struct Node *head, *temp, *prev;
    head = (struct Node *)malloc(sizeof(struct Node));
    head->data = 10;
    head->next = (struct Node *)malloc(sizeof(struct Node));
    head->next->data = 20;
    head->next->next = (struct Node *)malloc(sizeof(struct Node));
    head->next->next->data = 30;
```

```
    head->next->next->next = NULL;
    printf("Before Deletion:\n");
    display(head);
    temp = head;
    while(temp->next != NULL)
    {
        prev = temp;
        temp = temp->next;
    }
    prev->next = NULL;
    free(temp);
    printf("\nAfter Deletion at End:\n");
    display(head);
    return 0;
}
```

Output

```
Before Deletion:
10 -> 20 -> 30 -> NULL

After Deletion at End:
10 -> 20 -> NULL


=== Code Execution Successful ===
```

c) Write a C program for deletion of a newnode at any position of a single linked list .

```
#include <stdio.h>
#include <stdlib.h>
struct Node
{
    int data;
    struct Node *next;
};
void display(struct Node *head)
{
    struct Node *temp = head;
    while(temp != NULL)
    {
        printf("%d -> ", temp->data);
        temp = temp->next;
    }
    printf("NULL\n");
}
int main()
{
    struct Node *head, *temp, *prev;
    int pos = 3, i;
    head = (struct Node *)malloc(sizeof(struct Node));
    head->data = 10;
    head->next = (struct Node *)malloc(sizeof(struct Node));
    head->next->data = 20;
    head->next->next = (struct Node *)malloc(sizeof(struct Node));
```

```
    head->next->next->data = 30;
    head->next->next->next = (struct Node *)malloc(sizeof(struct Node));
    head->next->next->next->data = 40;
    head->next->next->next->next = NULL;
    printf("Before Deletion:\n");
    display(head);
    temp = head;
    for(i = 1; i < pos; i++)
    {
        prev = temp;
        temp = temp->next;
    }
    prev->next = temp->next;
    free(temp);
    printf("\nAfter Deletion at Position %d:\n", pos);
    display(head);
    return 0;
}
```

Output

```
Before Deletion:
10 -> 20 -> 30 -> 40 -> NULL

After Deletion at Position 3:
10 -> 20 -> 40 -> NULL


=== Code Execution Successful ===
```